

2. Central Nervous System - Structure and Function

General Objectives

- Describe the major structures and functions of the nervous system, and the features of nerve cells which adapt them to function as a communication system

Content

The major divisions of the nervous system. The somatic and autonomic divisions of the peripheral nervous system, their roles and interrelationships, especially the relationship between the control centres of the medulla and the autonomic nervous system.

The CNS, parts of the brain – cerebrum, cerebellum, hypothalamus, medulla, meninges – and their functions; the spinal cord and cerebrospinal fluid.

Parts of the brain and nerve pathways (names of nerves not required) in relation to the voluntary control of locomotion and object manipulation, and involuntary control of posture and balance.

1. 1999 / 13

The part of the brain associated with thought and reasoning is the

- (a) cerebellum.
- (b) medulla oblongata.
- (c) cerebrum.
- (d) hypothalamus.

3. 1999 / 29

The major control centres for respiration **and** heart rate are located in the

- (a) pons.
- (b) cerebellum.
- (c) pituitary.
- (d) medulla oblongata.

4. 2000 / 15

Which of this list is **NOT** a part of the body's peripheral nervous system?

- (a) Cranial nerves
- (b) Spinal nerves
- (c) Spinal cord
- (d) Touch receptor

5. 2000 / 16

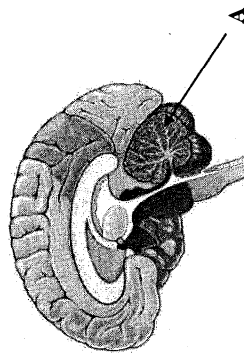
Name the part of the brain that regulates body temperature, hunger, thirst, and sleep.

- (a) Medulla
- (b) Hypothalamus
- (c) Cerebrum
- (d) Pons

7. 2001 / 6

Damage to structure A in the figure that follows of the brain would result in

- (a) uncoordinated movement.
- (b) loss of memory.
- (c) inability to regulate breathing.
- (d) impaired hearing.



8. 2002 / 11

The cerebellum does NOT control

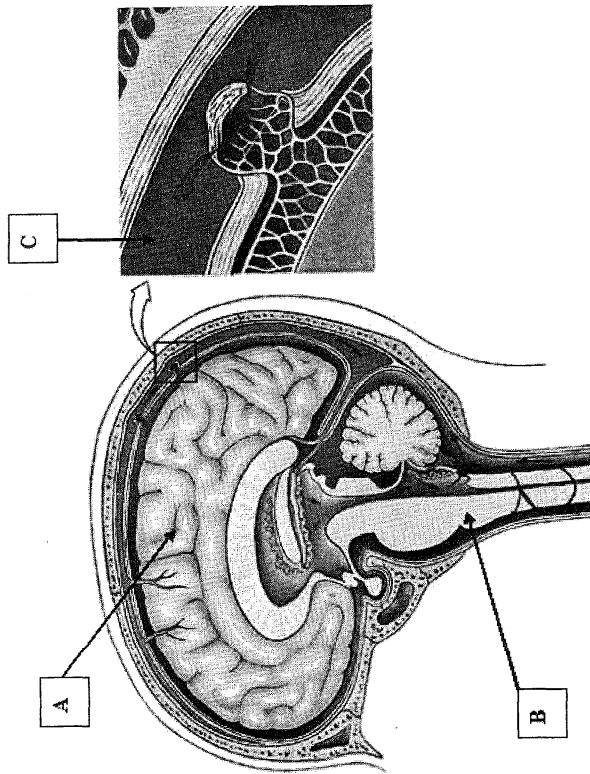
- (a) posture.
- (b) coordination.
- (c) balance.
- (d) body temperature.

9. 2003 / 12

Which of the following is an example of an effector in a reflex arc?

- (a) Pain receptor in the foot
- (b) Motor nerve fibres in the spinal cord
- (c) Motor end plate
- (d) Skeletal muscle

The next two questions refer to the diagram below.



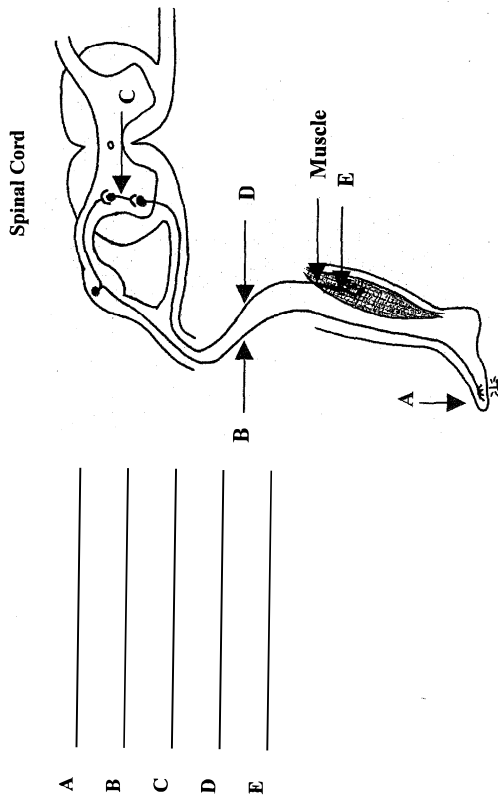
12. 2004 / 1

Which of the following is NOT true of structure A?

- (a) It is surrounded by three layers of meninges.
- (b) Information from the body terminates in structure A's white matter.
- (c) Its surface is convoluted to provide greater surface area.
- (d) It is connected to, and able to influence, the cerebellum.

15. 1999 / 49

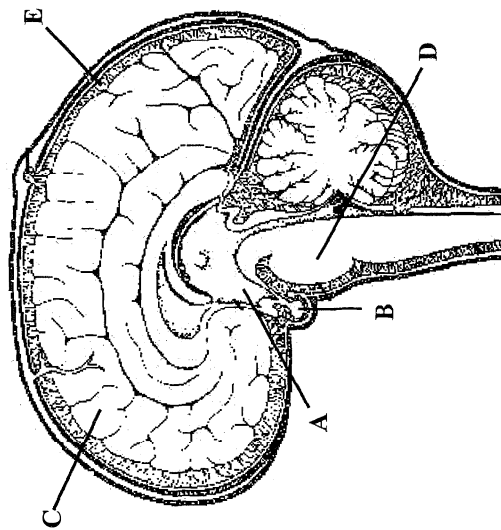
A spinal reflex allows us to withdraw quickly from a painful stimulus. Use the diagram below of a foot standing on a nail, to help you name the structures that make up a spinal reflex arc. (5)



16. 2000 / 49

(c) Describe why death could result if a small area of the medulla oblongata is damaged, while similar damage to the cerebrum may not be fatal. (2)

17. 2002 / 42



(a) Identify the labelled structures

- A _____ (1)
 B _____ (1)
 C _____ (1)
 D _____ (1)

(b) What is contained in the space labelled E?

- _____ (1)

EXTENDED ANSWERS

18. 2002 / 49

(b) An individual's spinal cord is severed below the level of the shoulder in a car accident. When examined, it is noticed that the patient's leg moves in response to a sharp stimulus to the foot. Explain how this response can occur. Include in your answer details of specific components of the pathway involved in this response. (7)

19. 2003 / 49

(b) Hospital patients who are in a coma experience a loss of function in many parts of their cerebrum and yet their heart continues to function normally.

Give an explanation for how the body is able to achieve this in the absence of a fully functioning cerebrum. (7)

3. Nervous System - Receptor Organs

General Objectives

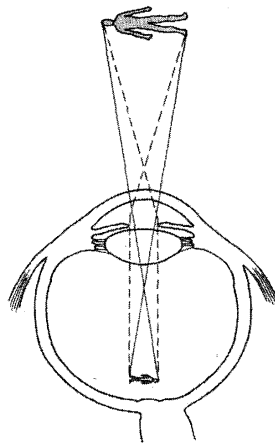
- Describe the major structures and functions of the nervous system, and the features of nerve cells which adapt them to function as a communication system

Content

The existence of receptors for external and internal stimuli, and the role of the senses such as hearing, balance and vision. Structure, function and defects of the eye and ear.

THE EYE

- 1999 / 14
One function of the cornea of the eye is to
 - convert light into nervous impulses.
 - allow light to enter the eye.
 - regulate the amount of light entering the eye.
 - maintain the shape of the eyeball.

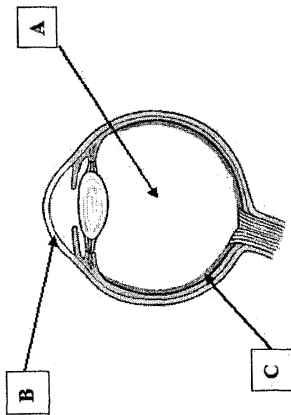


- (a) Each retina has about 7 million rods and 120 million cones.
 - (b) Cones provide black and white vision in bright light.
 - (c) Cones are found in the greatest concentration in the fovea centralis.
 - (d) Rods aid in colour vision when the light is dim.
- 2002 / 12
Which statement about the retina is correct?
 - (a) The sclera separates the eyeball into two cavities.
 - (b) The lens is held in position by the ciliary body.
 - (c) Adjustment of muscles in the iris changes the shape of the lens.
 - (d) Light entering the eye is reflected by both the aqueous and vitreous humour.
 - 2003 / 28
Which statement about the eye is **CORRECT**?
 - (a) The sclera separates the eyeball into two cavities.
 - (b) The lens is held in position by the ciliary body.
 - (c) Adjustment of muscles in the iris changes the shape of the lens.
 - (d) Light entering the eye is reflected by both the aqueous and vitreous humour.

2003 / 29

- In order to maintain focus on an object as it approaches more closely, the eye will
- contract the ciliary muscle causing the lens to bulge.
 - relax the ciliary muscle causing the lens to bulge.
 - tighten the suspensory ligaments on the lens and thereby flatten the lens.
 - dilate the pupil so that more light can be used to accommodate the object.

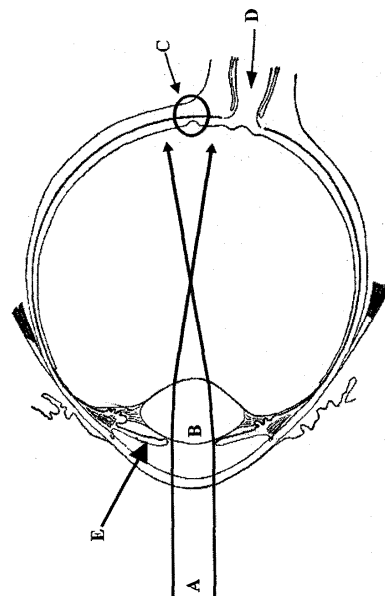
The next two questions refer to the diagram below.



- 2004 / 17
The structures labelled A, B and C in the diagram above are respectively
 - (a) aqueous humor, conjunctiva and sclera.
 - (b) vitreous humor, cornea and retina.
 - (c) vitreous humor, cornea and sclera.
 - (d) sclera, conjunctiva and choroid.

2001 / 41

- The diagram below shows a human eyeball in longitudinal section.



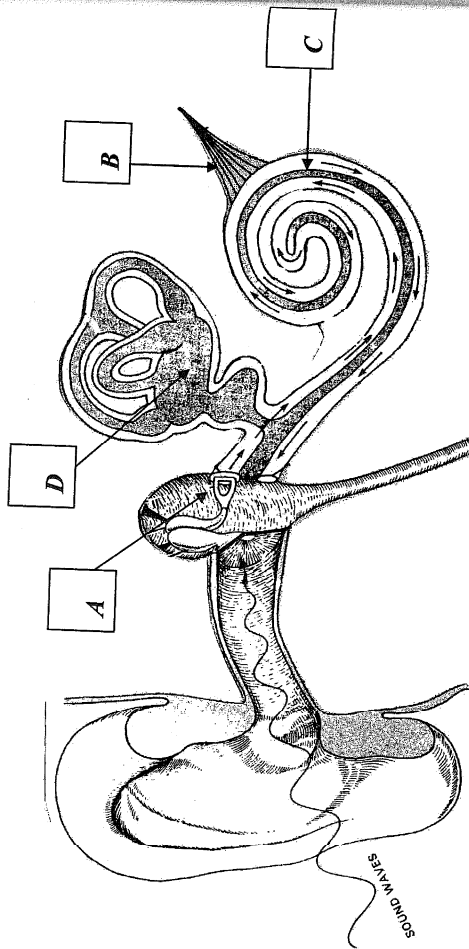
(2001 / 41 cont)

- (a) The arrows starting at A indicate a possible light pathway through the eyeball. What type of sight defect is being illustrated here?
- (b) What is structure B?
- (c) Apart from B, name **TWO** other components of the eye that bend (refract) the light passing into it.
- (d) What specific region of the retina is the arrow pointing to at C (circled)?
- (e) What type of photoreceptor is present in the highest concentration in C?
- (f) What is special about this region of the retina for normal human vision?
- (g) Name structure D.
- (h) What is the function of structure D?
- (i) Name structure E.
- (j) Describe the effect that structure E has on the pupil in response to bright and to dark light conditions.

THE EAR

9. 1999 / 16
The part of the ear that detects head position is the
- (a) auditory ossicles.
(b) tympanic membrane.
(c) eustachian (auditory) tube.
(d) semicircular canals.
10. 2000 / 17
Sensory hair cells can be found in all of the structures listed below **except** the
- (a) auditory canal.
(b) cochlea.
(c) semicircular canals.
(d) vestibule.
11. 2001 / 5
The ossicles are the major structures of the
- (a) cochlea.
(b) middle ear.
(c) inner ear.
(d) semicircular canals.
12. 2001 / 8
Receptors for hearing are located in the
- (a) tympanic membrane.
(b) semicircular canals.
(c) vestibule.
(d) cochlea.
13. 2002 / 23
A person is able to hear noises of a high frequency but is unable to hear noises of a low frequency. The most likely area of ear damage is the
- (a) oval window.
(b) semicircular canals.
(c) ossicles.
(d) cochlea.

The next two questions refer to the diagram below.



14. 2004 / 31

After stepping off an amusement park ride which involves spinning around, a person may experience dizziness because

- (a) the otoliths in structure D continue to move around.
- (b) the fluid and sensory hairs in structure C are disturbed by the spinning motion.
- (c) structure A can be pulled free from the inner ear.
- (d) structure B overstimulates the cerebrum.

15. 2004 / 32

Factory workers who have spent many years working next to loud machinery will be able to hear low frequency sounds but will experience some difficulty hearing sounds that are at a high frequency. This is because

- (a) the loud noises may have permanently fractured structure A.
- (b) some of the sensory hairs in structure C have been damaged.
- (c) loud noises can scar the sensory cells that line structure B.
- (d) scar tissue in structure D may interfere with its normal function.

16. 2000 / 49

Sensory impulses are physiologically identical regardless of the type of receptor involved. However, they give us completely different sensations. State briefly how this is accomplished.

_____ (1)

(b) Answer this part with reference to the diagram of an ear, below.

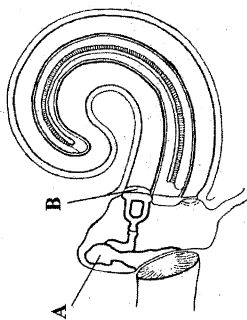


Diagram adapted from
Newton and Joyce "Human
Perspectives" 2nd Edition
1990.

(i) Describe the function of structure A.

_____ (1)

(ii) If structure B were to harden and become rigid, describe the effect this could have on the hearing process.

_____ (1)

EXTENDED ANSWERS

17. 1999 / 51

The body uses many different types of receptors to sense its environment. Through special senses such as vision and hearing, we can detect very complicated information. Describe how receptors in the eye and ear receive and process variable light and sound information, and communicate this to the brain.

(20)

18. 2003 / 50

(a) A major form of deafness is conductive deafness. Define this term and describe its causes.

(5)

(b) Describe how normal balance, posture and locomotion are controlled by the human body.

(15)

4. Nervous System - Neurons

General Objectives

- Describe the major structures and functions of the nervous system, and the features of nerve cells which adapt them to function as a communication system

Content

The neuron as the structural and functional unit of the nervous system – cell body, dendrites, axon, myelin sheath, nodes of Ranvier, motor end plate. Structure and function of receptor, connector and effector neurons.

1. 2000 / 14

In a neuron, the impulse is conducted through the dendrite to the

- myelin sheath.
- axon.
- associated sensory neuron.
- cell body.

The next two questions refer to the diagram below.

4. 2003 / 18

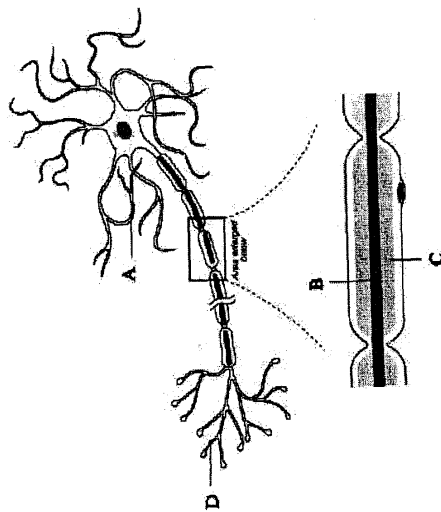
Which part of the structure shown above increases the speed of nerve impulse transmission?

- A
- B
- C
- D

5. 2003 / 19

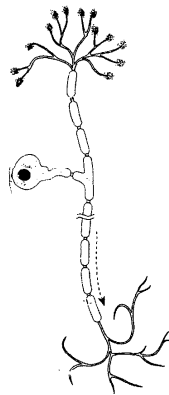
Which one of the following statements about the neuron that follows is correct?

- This neuron is a sensory neuron.
- Structure B will produce the myelin sheath.
- Structure D receives impulses from other neurons.
- Structure A transmits impulses to the cell body.



6. 2004 / 6

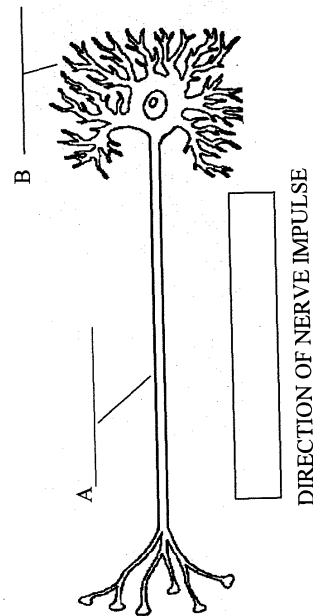
The neuron illustrated below would be classified correctly as



- a sensory neuron.
- a motor neuron.
- a connector neuron.
- there is not enough information to say.

7. 2000 / 49

Label the structures A and B of the neuron drawn below. Draw an arrow in the box provided to indicate the direction of the nerve impulse.



27. Scientific Method

General Aims

To develop competence in the:

- Process skills of science associated with designing and performing controlled experiments, collecting, recording, presenting and interpreting data
- Measurement of length, volume and mass, and in the calculation of magnifications, field diameters and estimations of dimension of structures in light

General Objectives

- Demonstrate competence in measurement of length, volume and mass, and in calculations of magnifications, field diameters and estimations of dimension of structures in light microscopy.
- Retrieve information from a variety of sources, collate information into succinct reports, communicate accurately and clearly, both orally and in writing
- Analyse and interpret data presented in graphical and tabular form

The next two questions refer to the table below.

Consider the table below showing data recorded for a person viewing a frightening horror movie.

Time (minutes)	1	2	3	4
Cardiac Output (L / minute)	5	7	20	10

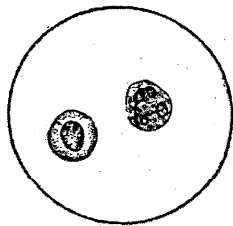
1. 2003 / 26

Concerning these data, which of the statements below is most correct?

- The sympathetic nervous system appears to be most active between 2 and 3 minutes.
- The parasympathetic nervous system is most active at 3 minutes.
- Adrenalin is secreted in response to increased cardiac output at 4 minutes.
- Between 2 and 3 minutes the hypothalamus adjusts the heart rate to meet a need for greater oxygen.

3. 2003 / 32

Cells shown in the diagram below were observed using a microscope fitted with a 40X objective lens and a 10X ocular (eyepiece) lens. Given that the field of view for this microscope at a magnification of 100X is 200 micrometres, what is the approximate size of these cells?



- 10 micrometres
- 20 micrometres
- 30 micrometres
- 50 micrometres

4. 2003 / 33

If human cells were given essential nutrients and allowed to grow and reproduce in a closed test tube, what changes would you expect to measure in the test tube?

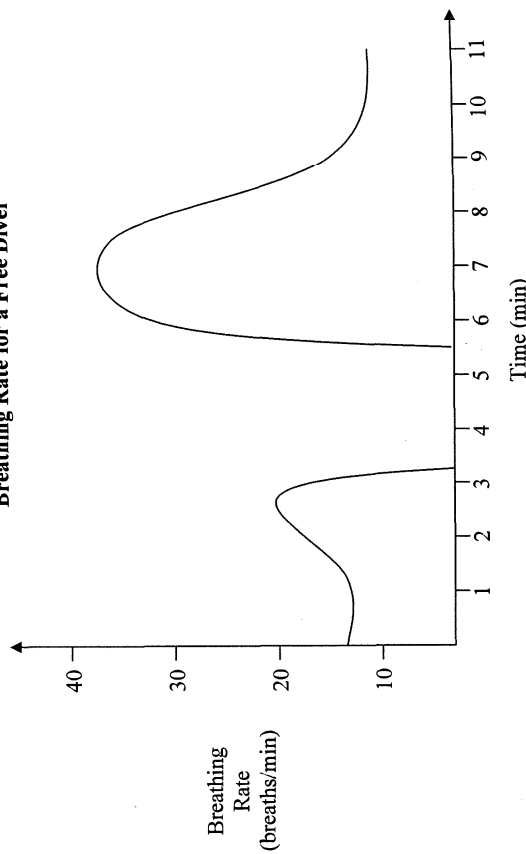
- An increase in temperature, as well as increased levels of oxygen and carbon dioxide.
- An increase in temperature, carbon dioxide levels and glucose but decreased oxygen levels.
- An increase in temperature and carbon dioxide levels but decreased levels of oxygen and glucose.
- A decrease in temperature as well as a decrease in levels of oxygen, carbon dioxide and glucose.

5. 2004 / 25

Free-diving is an Extreme sport in which divers hold their breath for long periods of time and dive to great depths. An investigation into the effect that such a dive has on breathing rate was performed.

The data collected for breathing rate are shown below.

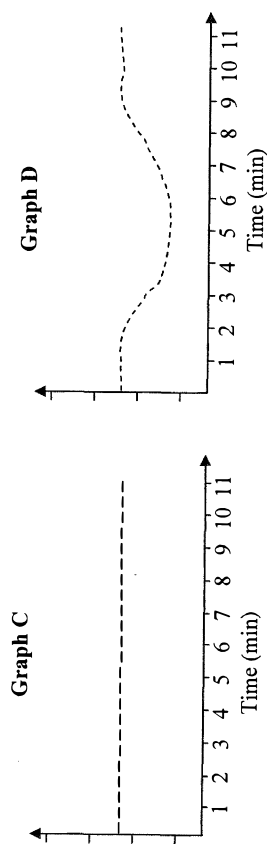
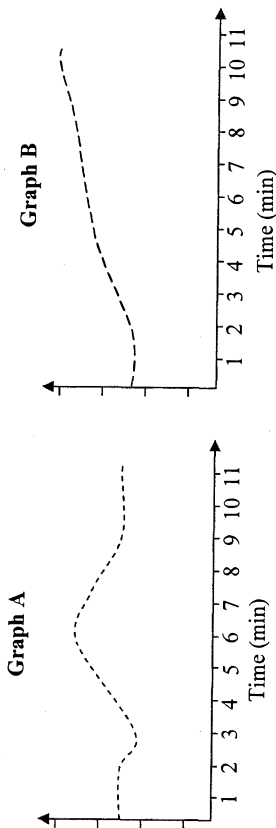
Breathing Rate for a Free Diver



Examine the graphs drawn below. Which of the graphs best represents the change in carbon dioxide levels in the blood during the free dive investigation?

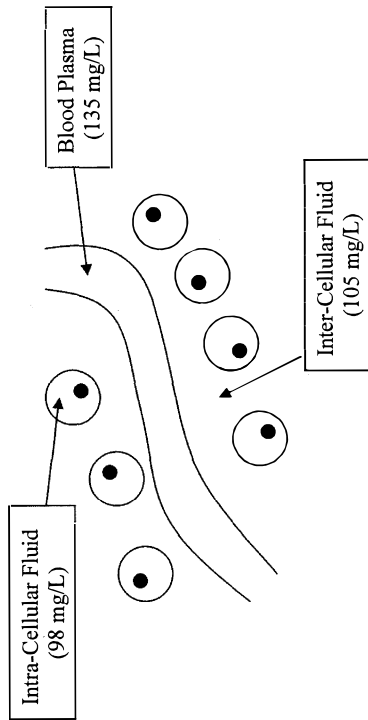
- (a) Graph A
- (b) Graph B
- (c) Graph C
- (d) Graph D

Note: The Y-axis represents levels of Carbon Dioxide in the blood.



2004 / 35

The diagram below indicates part of an image viewed down a microscope. Three different fluid compartments within a small sample of human tissue are shown. Also provided are data on the concentration of total dissolved substances within these compartments.



If you were asked to measure the size of these cells, which piece of information would be most useful?

- (a) The field of view.
- (b) The magnification of the image.
- (c) The number and size of lenses in place.
- (d) The type of cells being viewed.

1999 / 45

A drug company has developed a new influenza vaccine that is squirted into the nasal passages. The vaccine has been approved for testing in people, to see if it prevents the symptoms of influenza in people at risk of infection. Consider how an experiment might be designed to test this vaccine, and answer the following questions.

- (a) State a suitable **hypothesis** for this experiment. (1)
- (b) What would be the **independent variable** in the experiment? (1)
- (c) What would be the **dependent variable** in the experiment? (1)

(1999 / 45 cont)

(e) List two variables that would need to be controlled in this experiment.

(i)

(ii)

(2)

(f)

(2002 / 45 cont)

TREATMENT	Tube number	GROWTH OF NERVE AXONS AFTER ONE MONTH (mm)
NEUROGEN (Large, single dose)	1	0.50
	2	0.42
	3	0.45
NEUROGEN (Series of small doses)	4	0.80
	5	0.98
	6	0.90
NO TREATMENT	7	0.25
	8	0.27
	9	0.19

For the experiment described above:

(a) Identify **ONE** hypothesis being tested. (2)

(b) What is the Dependent Variable? (1)

(c) Based on the information given, list **three** variables that were controlled in this experiment.

(i)

(ii)

(iii)

(3)

(d) What is the purpose of having the three test tubes that did not receive any treatment with Neurogen? (2)

(e) What **two** major conclusions could be drawn from these results? (2)

(i)

(ii)

(f) After this experiment, the scientist repeated the procedure three more times. Why would the scientist repeat the experiment so many times? (2)

8.

(c)

9. 2002 / 45

A scientist was investigating the effectiveness of a newly developed drug, "Neurogen", and its ability to improve recovery from nerve cell injury. The experiment was performed as described below:

Nerve cells were placed into culture medium (a solution containing nutrients to keep the cells alive and help them grow). An equal number of nerve cells (in culture medium) were transferred into nine test tubes and maintained under constant temperature and oxygen levels. Three of the test tubes received no further treatment. Of the remaining six test tubes, three tubes received a large, single dose of Neurogen whilst the remaining three tubes had a series of smaller doses of Neurogen added over a one week period.

After one month the growth of nerve axons in the test tubes was measured and the results are shown in the table below.

(2)